Questions

Q1.

(i) State the equation that relates acceleration to change in velocity and time taken.

(1)

(ii) A van accelerates from a velocity of 2 m / s to a velocity of 20 m / s in 12 s.
Calculate the acceleration of the van.

(2)

acceleration = ... m / s^2

(Total for question = 3 marks)

Q2.

A different car has a device that can detect rain.

This device is linked to a computer that can change the speed of the car.

In wet weather, the computer changes the speed of the car.

(i) State the change in speed that the computer should make when the road is wet.

(1)

.....

.....

(ii) Give a reason why this change in speed is necessary when the road is wet.

(1)

.....

.....

(Total for question = 2 marks)

Q3.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A car is being driven at a constant velocity.

The driver sees an obstacle in the road ahead.

The driver uses the brakes to stop as quickly as possible.

Figure 9 shows the velocity/time graph for the car from the time when the driver sees the obstacle.

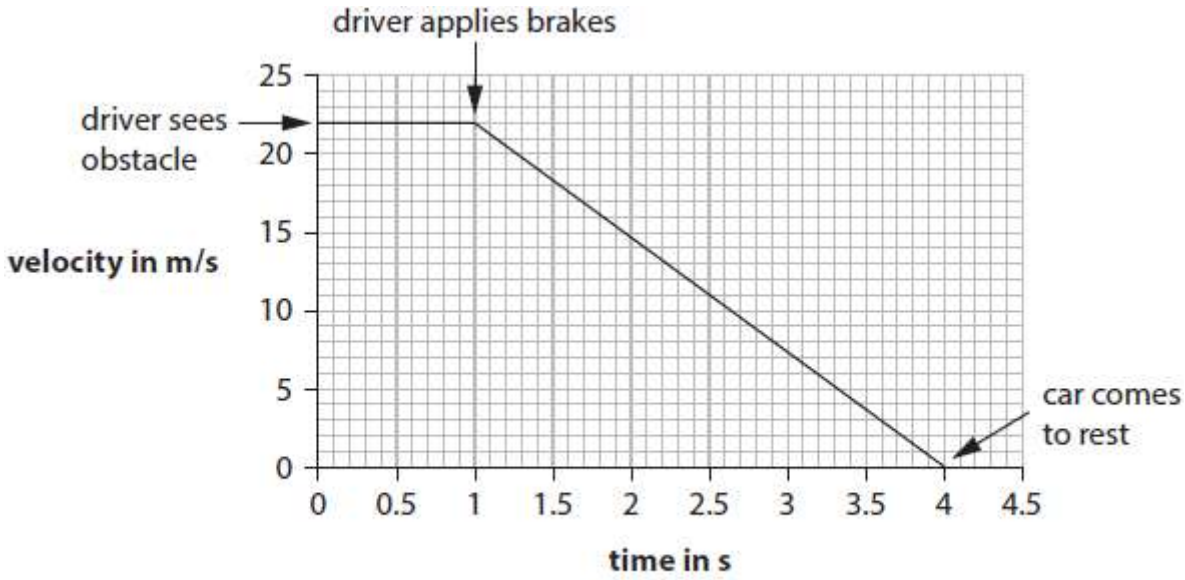


Figure 9

(i) Which of these is the driver's reaction time shown in Figure 9?

(1)

☐ **A** 0 s
☐ **B** 1 s
☐ **C** 4 s
☐ **D** 22 s

(ii) State **one** factor that might increase a driver's reaction time.

(1)

.....

.....

(iii) Calculate the distance travelled between when the driver applies the brakes and when the car comes to rest in Figure 9.

Use the equation

distance = area under the sloping line of the graph in Figure 9

(3)

distance = ... m

(Total for question = 5 marks)

Q4.

Two people, L and M, have a 100 m race.

L starts running before M.

Figure 20 shows a distance/time graph of the race.

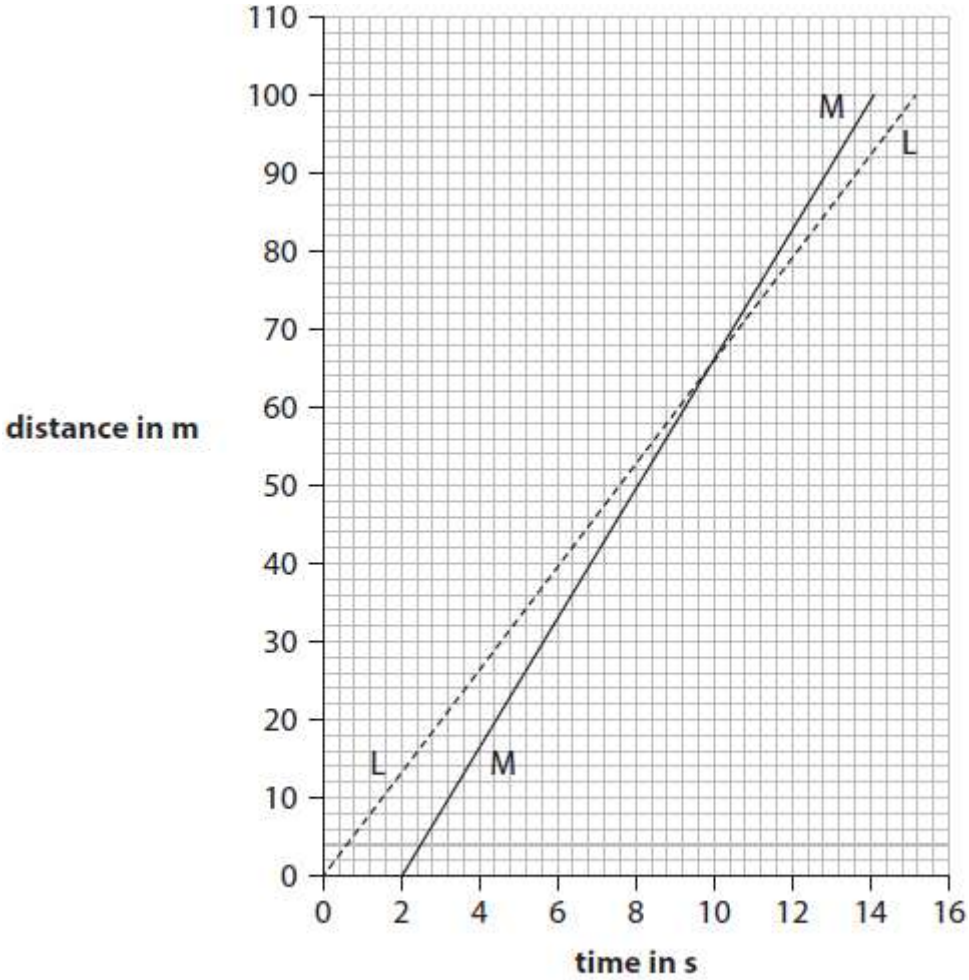


Figure 20

(i) State the **distance** that L has run when M overtakes.

(1)

distance = ... m

(ii) Calculate the velocity of L when running the 100 m race.

(2)

velocity = ... m/s

(Total for question = 3 marks)

Q5.

Which of these speeds would be normal for a person walking?

(1)

- ☐ **A** 0.1 m/s
- ☐ **B** 1.0 m/s

☐ **C** 10 m/s
☐ **D** 100 m/s

(Total for question = 1 mark)

Q6.

Some students investigate the speed of cars.
They measure the time it takes each car to travel a distance of 80 m.

(a) State **two** measuring instruments the students should use.

(2)

1

.....

2

.....

(b) The table shows some of their results.

colour of car	distance travelled / m	time / s
green	80	5.0
red	80	4.0
blue	80	5.5
black	80	4.3
white	80	5.6

(i) State the colour of the slowest car.

(1)

colour of the slowest car .....

(ii) Calculate the speed of the black car.

(2)

speed of the black car = ... m/s

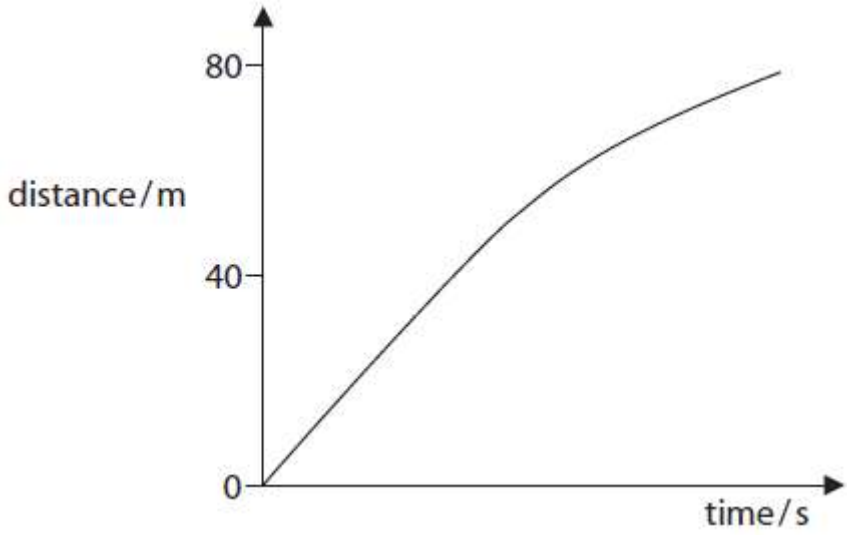
(iii) 20 miles per hour is approximately 9 m/s.
Estimate the speed, in miles per hour, of the black car.

(1)

speed of the black car = ... miles per hour

Q7.

The distance-time graph for a car is shown below.



Describe what the graph shows about the speed of the car as it travels the 80 m.

(2)

.....

.....

.....

.....

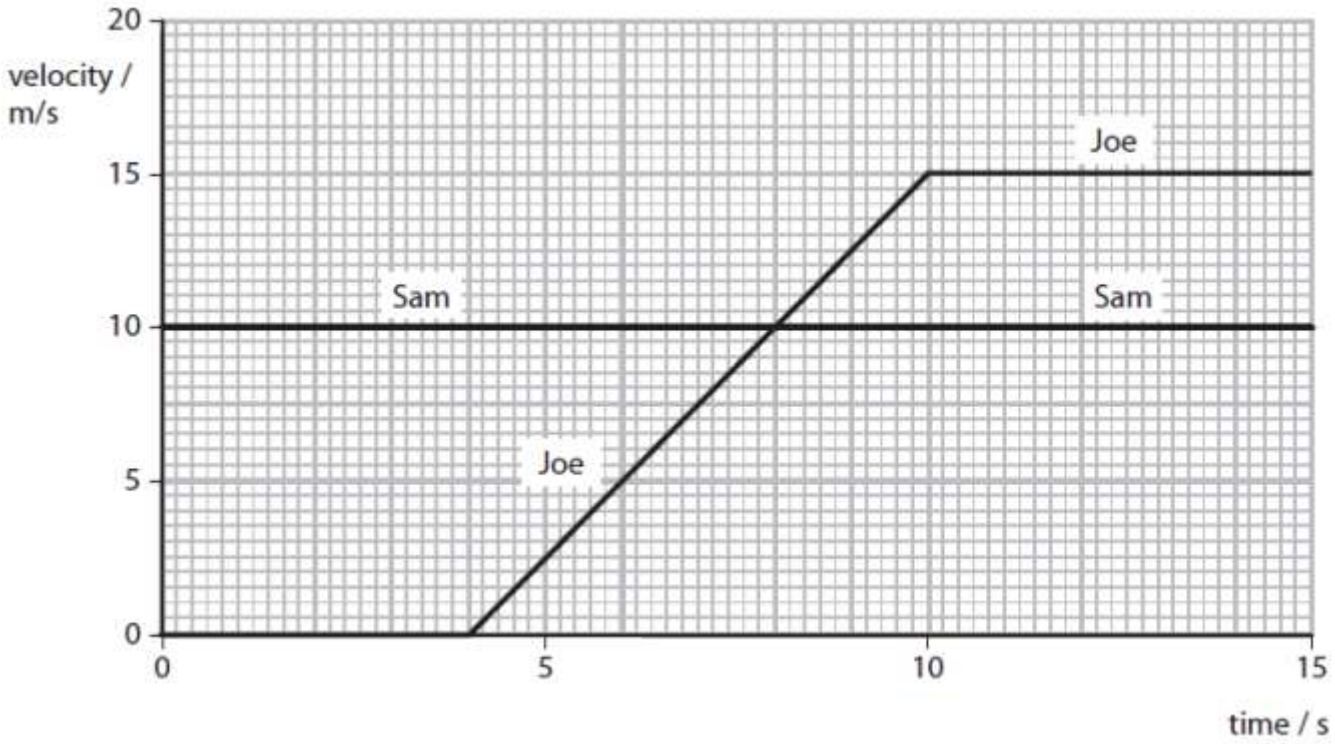
.....

Q8.

Sam and Joe are on their bikes. They are on a flat, straight road.

(a) Joe is stationary when Sam rides past at a constant velocity of 10 m/s.

Joe waits for 4 s and then follows Sam.
This is a velocity/time graph of their motion.



(i) How far did Sam travel during these 15 s?

distance = velocity × time

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** 1.5 m
☐ **B** 10 m
☐ **C** 100 m
☐ **D** 150 m

(ii) At which of these times is the resultant force on Joe bigger than the resultant force on Sam?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** at 3 s
☐ **B** at 7 s
☐ **C** at 11 s
☐ **D** at 15 s

(iii) For how many seconds was Joe accelerating?

(1)

number of seconds = ... s

(iv) Calculate Joe's acceleration during this time.

(2)

Joe's acceleration = ... m/s^2

(b) The diagram shows the horizontal forces acting on Joe at one point while he is accelerating.

(i) Calculate the size of the resultant horizontal force acting on Joe and his bike.

(2)

size of resultant force = ... N

(ii) The total mass of Joe, his heavy bag, and his bike is 55 kg.
Calculate the total weight.
Gravitational field strength, g = 10 N/kg

(1)

total weight = ... N

(c) On another day, Joe is riding the same bike on the same piece of road.

This time he does not have the heavy bag on his back.
He finds that it is easier to accelerate.
Explain why Joe finds it easier to accelerate.

(2)

.....

.....

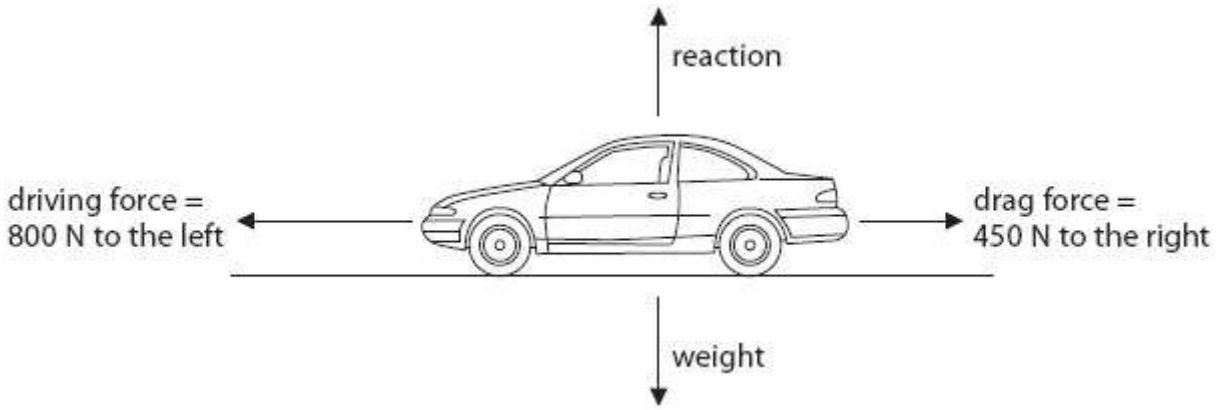
.....

.....

(Total for Question = 10 marks)

Q9.

The diagram shows the forces acting on a car which is travelling along a flat straight road.



(a) (i) The size of the resultant force on the car is 350 N.
In which direction is the resultant force acting?
Put a cross (■) in the box next to your answer.

(1)

☒ **A** down ↓

☒ **B** to the left ←

☒ **C** to the right →

☒ **D** up ↑

(ii) Complete the sentence by putting a cross (■) in the box next to your answer.
The car is

(1)

☒ **A** accelerating

☒ **B** decelerating

☒ **C** moving at a constant speed

☒ **D** not moving

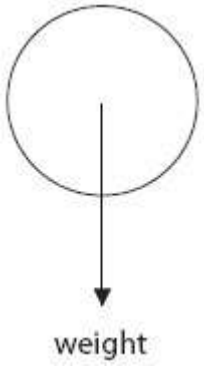
(2)

(iii) The mass of the car is 625 kg.
Calculate the weight of the car.
gravitational field strength = 10N/kg

(2)

.....

(b) Forces also act on objects when they fall through the air.
There are two forces acting on this ball as it falls through the air.
The weight is shown on the diagram.



(i) Draw and label an arrow on the diagram to show the other force acting on the ball.

(2)

(ii) Use words from the box to complete the sentences.

(2)

balanced	changing	greater	smaller	zero

After a short time the ball falls at a steady speed.

The forces acting on the ball are now .....

The acceleration of the ball is now .....

(Total for Question is 8 marks)

Q10.

Figure 1 shows a block hanging from a spring balance.

Figure 1

Use a word from the box to complete the sentence below.

density	mass	volume	weight

The quantity measured by the spring balance in Figure 1 is

(1)

.....

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	acceleration = $\frac{\text{change in velocity}}{\text{time (taken)}}$	$a = \frac{v-u}{t}$ $a = \frac{\Delta v}{t}$ $\frac{v}{t}$ allow correct rearrangements seen here or in bii	(1)

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) $\frac{20 - 2}{12}$ evaluation (1) 1.5 (m/s²)	$\frac{18}{12}$ -1.5 (m/s²) award full marks (1 in bi and 2 in bii) for the correct answer without working, award 1 mark if 20-2 or 18 or 2-20 is seen and no other marks are scored If (incorrectly) $a = \frac{v^2 - u^2}{t}$ given in 3bi $a = \frac{20^2 - 2^2}{12}$ OR = 33 scores 1 mark	(2)

Q2.

Question number	Answer	Additional guidance	Mark
(i)	speed should be reduced (1)	slower accept specified numerical reduction e.g. from 40 to 20	**1** **AO3.1**

Question number	Answer	Additional guidance	Mark
(ii)	any **one** from friction (between wheel and road) is reduced (1) braking distance would be longer (on wet road) (1)	allow reverse argument idea that (road) more slippery / car could skid less traction	**1** **AO3.1**
	(overall) stopping distance would be longer (on wet road) (1)	longer (time/distance) to stop ignore crash / collision/ accident ignore changes to thinking distance	

Q3.

Question number	Answer	Additional guidance	Mark
(i)	B 1s A is incorrect because this is before the driver reacts. C is incorrect because this is the braking time D is incorrect because 22 is a value of the initial velocity		**1** **AO2.1**

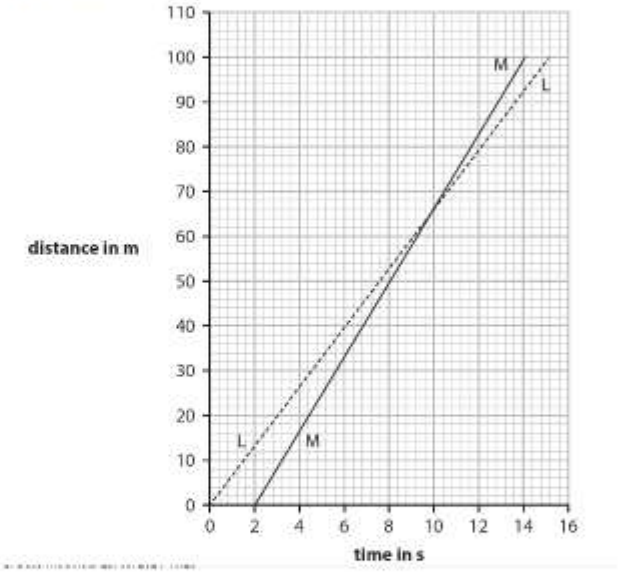
Question number	Answer	Additional guidance	Mark
(ii)	Any **one** from • tiredness • distraction • drugs / alcohol • (old) age	accept similar descriptions any named drug	**1** **AO1.1**

Question number	Answer	Additional guidance	Mark
(iii)	attempt to use the correct part of the graph (1) attempt to find area under graph(1) ½ x b x h evaluation (1) 33 (m)	values of 22(±0.5) and 4 or 22(±0.5) and 3 seen 0.5 x 4 x 22(±0.5) or 0.5 x 3 x 22(±0.5) accept values between 32 and 34 for 3 marks accept values between 43 and 45 for 2 marks accept values between 64.5 and	**3** **AO2.1**

		67.5 for 1 mark. accept values between 86 and 90 for 1 mark award full marks for the correct answer without working	

Q4.

Question number	Answer	Additional guidance	Mark
(i)	66 (m)	allow values between 64 and 68 inclusive allow values between 32 and 36 as the distance L has to run after M overtakes in this context	**1** **AO3.2**

Question number	Answer	Additional guidance	Mark
(ii)	select (1) $v = \frac{x}{t}$	allow any identifiable distance from graph divided by any identifiable time from graph e.g. $\frac{100}{15.2}$  	**2** **AO2.1**
	evaluation (1) 6.6 (m/s)	allow values that round to between 6.5 (m/s) and 6.7 (m/s) for example 6.666 (m/s) or 6.579 (m/s) award full marks for correct answer without working	

Q5.

Question Number	Answer	Mark
	B 1.0 m/s **The only correct answer is B** *A 0.1 m/s is incorrect, being 1 metre every 10s, insect crawling pace* *C 10 m/s is incorrect, being an Olympic sprinter's pace, much too fast for 'walking'* *D 100 m/s is incorrect, being a very fast sport's car's pace*	**(1)** AO 1 1

Q6.

Question Number	Answer	Acceptable answers	Mark
(a)	stopwatch /stopclock (1) {trundle/measuring} wheel/measuring tape or tape measure (1) ignore speedometer/speed camera/radar	(electronic) timer timing app (on 'phone) clock and watch on their own are insufficient any suitable length measuring device e.g. accept metre {rule(r)/stick} but ruler on its own is insufficient **Answers may be in either order**	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b) (i)	white (car) (1)	Allow the use of other columns that identify correct car e.g. 5.6(s)	**(1)**

Question Number	Answer	Acceptable answers	Mark
(b) (ii)	substitution (1) 80 ÷ 4.3 evaluation (1) 19 (m/s) Throughout the paper do not penalise answers to many places of decimal e.g. here 18.604651 gets both marks	Allow full marks for correct answer with no working seen. accept 18.6 (m/s) ignore 18 and 18.0 as incorrect rounding accept any power of 10 error for 1 mark	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b) (iii)	40 (miles per hour) (1)	accept answers in range 39 – 43 (miles per hour) ecf from b(ii) multiply bii by 2.222 range +/- 2.0	**(1)**

Q7.

Question Number	Answer	Acceptable answers	Mark
	{steady/constant} speed (at first) (1) (then) slows down (1)	accept velocity for speed ignore as time increases distance travelled increases (then) slower/less speed/decelerates/negative acceleration	**(2)**

Q8.

Question Number	Answer	Acceptable answers	Mark
(ai)	**D** 150 m **(1)**		**(1)**

Question Number	Answer	Acceptable answers	Mark
(aii)	**B** at 7 s **(1)**		**(1)**

Question Number	Answer	Acceptable answers	Mark
(aiii)	6 (s) **(1)**		**(1)**

Question Number	Answer	Acceptable answers	Mark
(aiv)	Substitution: 15 ÷ 6 (1) Evaluation 2.5 (m/s^2) (1)	Allow ecf from 4(aiii) Must be 15 divided by their 4(aiii) ECF allowed from first marking point ie evaluation of 15 divided by their answer from 4(aiii) Award 2 marks for correct answer, no working	**(2)**

Question Number	Answer	Acceptable answers	Mark
(bi)	100 - 30 (1) 70 (N) (1)	100 + 30 or 130 gains 1 mark Award 2 marks for correct answer, no working	**(2)**

Question Number	Answer	Acceptable answers	Mark
(bii)	550 (N) (1)	539 (N) allow use of g = 9.8 N/kg 539.55 (N) for use of g = 9.81N/kg Award mark for correct answer, no working	**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)	An explanation linking (combined) mass is less (1) smaller force required for same acceleration OR more acceleration from same force (1)	ignore references to weight, friction or backwards force ignore "easier to accelerate" as in stem less force needed (to accelerate)	**(2)**

(Total for Question =10 marks)

Q9.

	Answer	**Acceptable answers**	**Mark**
(a)(i)	B to the left ←		**(1)**
(a)(ii)	A accelerating		**(1)**
(a)(iii)	substitution 625x 10 (1) Evaluation 6250 (N) (1)	625 x 9.8 6125 (N) give full marks for correct answer, no working	**(2)**
(b)(i)	↑ (1) air resistance (1)	upward arrow on any part of line vertical line from any point on the diagram air friction, upthrust, drag Ignore any downward arrow labelled weight or gravity	**(2)**

(b)(ii)	Balanced (1) Zero (1)		**(2)**

Total for marks for question = 8

Q10.

Question Number	Answer	Mark
	weight / force (accept circle around weight if not contradicted on answer line)	**(1)** AO 1 2